

Hydraulic System Maintenance & Failure Response

Standard Operating Procedure

Doc ID	SOP-MNT-001
Version	2.0
Effective Date	2026-02-01
Owner	Maintenance Department
Scope	Press Line units (Plant A, Plant C); applies to all hydraulic power packs, cylinders, hoses and seals

1. Purpose

Standardizes routine hydraulic maintenance and the emergency response to a DT-01 hydraulic system failure, which is the highest-severity and longest-duration downtime code recorded plant-wide.

2. Required PPE

- Safety glasses with side shields
- Oil-resistant gloves
- Apron

3. Tools & Materials

- Pressure gauge kit
- Replacement seal kit
- ISO VG 46 hydraulic oil
- Torque wrench
- Oil absorbent pads

4. Procedure

- 1 Isolate hydraulic pump and relieve accumulator pressure to zero before opening any line.
- 2 Identify leak/failure source: hose, cylinder seal, pump, or valve block.
- 3 For hose failure: replace with like-for-like rated hose assembly; torque fittings to spec.
- 4 For cylinder seal failure: remove cylinder, replace seal kit, check bore for scoring before reassembly.
- 5 Refill reservoir to correct level with ISO VG 46 oil; bleed air from the circuit.
- 6 Restart pump at reduced pressure and check for leaks across 5 full cycles before returning to full rate.
- 7 Record failure mode, parts replaced, and total downtime in the incident log (cross-reference incidents/ folder).

5. Associated Downtime Code

Code	Description	Severity	Typical Duration
DT-01	Hydraulic System Failure	Major	137 min

■ Hydraulic fluid under pressure can penetrate skin. Always relieve system pressure to zero and verify with gauge before disconnecting any fitting.